

You can change run time parameters of the SELinux system by directly writing to the files in this pseudo file system. As an example, just issue this command as the root user:

```
[root@vbg ~]# echo 1 > /selinux/enforce
```

You will see that the mode of SELinux has changed from Permissive to Enforcing.

To return back to Permissive Mode, you can run either of the commands:

```
[root@vbg ~]# echo 0 > /selinux/enforce
```

or

```
[root@vbg ~]# setenforce 0
```

The third line mentions the current SELinux mode, whereas the fourth line mentions the SELinux mode under which the system booted.

The fifth line mentions the version number of the policy (we will come to this later in this series) and finally, the sixth line mentions the Policy loaded from the configuration file (*/etc/sysconfig/selinux*) at boot time.

Understanding various types of policies

By definition, an SELinux policy is a collection of rules for SELinux Mandatory Access Controls. Each one of us can make a policy to suit our needs much like we define firewall rules through *iptables*. There can be no standard policy that can apply to all situations.

By default, there are two policies shipped along with Red Hat Enterprise Linux: Targeted and Strict.

The Targeted Policy is the first step in assisting system administrators to understand and implement SELinux. It only 'targets' certain network daemons such as the Apache Web server, FTP server, BIND DNS server and a few others, while leaving the vast majority of end-user applications largely untouched. It creates an 'unconfined' domain 'confinement' (interesting paradox, isn't it?) and does not apply Access Control Restrictions to most applications in the unconfined domain.

This allows sysadmins to concentrate on the really vulnerable network applications and services while not interfering with their daily tasks.

Once the nuts and bolts of SELinux are clear to administrators, they should move forward towards implementing the SELinux Strict Policy.

The Strict Policy, on the other hand, is a true restrictive Access Control Policy. Before implementing this policy, make sure you understand SELinux concepts and policies well.

Under the hood of Targeted Policy

To understand any SELinux policy, use the informative command—*seinfo*.

```
[root@vbg ~]# seinfo
```

Statistics for policy file: /etc/selinux/targeted/policy/policy.21
Policy Version & Type: v21 (binary, MLS)

```
Classes:      61  Permissions:  220
Types:       1513 Attributes:  148
Users:        3  Roles:        6
Boolean:     210  Cond Expr:   186
Sensitivities: 1  Categories:  1024
Allow:       82518 Neverallow:   0
Auditallow:  28  Dontaudit:   5086
Role allow:   5  Role trans:   0
Type_trans:  1398 Type_change:  17
Type_member:  0  Range_trans:  23
Constraints:  47  ValidateTrans:  0
Fs_use:       15  Gendfscon:    64
Portcon:      264 Netifcon:    0
Nodecon:      8  Initial SIDs:  27
```

seinfo is a policy query tool that queries policy files and provides vital information about it. When executed without any arguments, it queries the default loaded policy file.

As you can see the policy file read by the *seinfo* tool by default is */etc/selinux/targeted/policy/policy.21*. Installed by the *selinux-policy-targeted* RPM, this file contains the binary Targeted Policy.

The second line of the above output mentions that this is a binary policy with MLS (Multi Level Security). We will come to MLS in the later part of this series.

Now comes the interesting part of the output. You can clearly see various components of the policy. We will discuss these components in the course of these articles. Let us concentrate on a few important ones right now.

A typical security context as we discussed in Part 1 of this series, is of the type:

```
User Identity:Role:Type/Domain
```

As we can see from the above, in the default SELinux Targeted policy, there are:

- Users (3 in number)
- Roles (6 in number)
- Types (1,513 in number)

This means that any object or subject in the SELinux Policy installed in the system can have one of three user identities, one of six roles and one of the available 1,513 types.

To list user identities defined in the SELinux Targeted Policy, run the following command:

```
[root@vbg ~]# seinfo -u
```

```
Users: 3
system_u
root
user_u
```

Do check all possible security contexts for objects (files,